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A Simple Tool to Estimate Transport GHGs Mitigated from Compact Urban Form

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Featured Application

The simple model allows for a preliminary assessment of the potential benefits from implementing compact urban form before investing in more sophisticated modeling. The model is best suited for car-centric municipalities from wealthier nations.

Abstract

Compact urban form can reduce road transportation GHG emissions and mitigate resource supply bottlenecks associated with mass EV adoption. Global databases from Climate TRACE and the Global Human Settlement Layer are utilized to develop the Compact Urban Form Estimation Tool or CUFET for calculating the reduction in VKT and road transportation GHGs from shifting toward CUF. The CUFET does not explicitly account for mechanistic changes in driving (e.g., modal shift) but rather uses settlement density as a coarse proxy for walking and transit urban fabrics. VKT was modeled using weighted least squares regression from the independent variables settlement population, settlement population density, and country fixed effects. Population size banding was introduced to the model to improve explanatory power. The model was developed using 10,495 settlements in the 2021 Climate TRACE dataset. The CUFET VKT model was able to explain 78% ($p < 0.001$) of the variation in the VKT of test settlements and improved with the addition of a country fixed effect. The CUFET on average gave estimates of VKT within 24% of Climate TRACE-calculated VKT for countries with a GDP per capita between $20,000 and $45,000 and average estimates within 20% for countries with a GDP per capita above $45,000. Increased settlement density was associated with more substantial reductions in VKT in small (50,000 to 88,335) and medium (88,335 to 329,480) sized settlements relative to large (>329,480) settlements. Higher variability was observed in VKT estimates of small settlements. The CUFET VKT was validated by backcasting historical VKT data from 1960 to 2000. The backcasting exercise used historical administrative boundaries and only included high economic output nations (GDP per capita above $20,000 in 2021 USD). Despite these limitations, backcasting achieved a % difference of ~20% for settlements after 1990, suggesting the model can make useful estimates within 30 years of the model calibration year for high economic output nations. The VKT model was used to calculate emissions using a settlement-specific emissions factor. Settlements with annual road transportation emissions per capita greater than 2 t CO_2eq have the lowest population densities relative to their populations and are mostly located in the United States, Japan, Canada, and Australia. The nations with the highest transportation emissions are also nations where the CUFET provides the most accurate VKT estimates. The CUFET aims to bridge the gap between academic consensus and local decision-making practice by

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reducing the barriers to estimate VKT and transportation GHG reduction from shifting to compact urban form.

Keywords: road transportation; GHG emissions; population density

1. Introduction

The world lags behind the decisive greenhouse gas (GHG) emission mitigation required to keep the global average temperature below a 2 °C increase from pre-industrial levels [1]. The transport sector contributed 16% of global GHG emissions in 2023 [2], with wealthy autocentric countries such as the United States, Canada, and Australia continuing to disproportionately contribute to transportation emissions [2,3].

Electric vehicles (EVs) will be an important technology to reduce emissions; however, treating EVs as a silver bullet to road transportation emissions may lead to missed mitigation targets [4]. Potential barriers to EVs replacing internal combustion engine (ICE) vehicles by 2050 include a risk of elevated electricity prices due to a range of supply-and-demand-related factors. Supply impediments include lagging transmission and distribution infrastructure development [5], while increased demand is expected from vehicle fleets transitioning to EVs [4], increased air conditioning in a warmer world [6], artificial intelligence (AI) data centers [7], and atmospheric CO_2 removal [8]. Additionally, supply bottlenecks of key metals and minerals may hinder the scale of EV production required to replace ICEs by 2050 [9–11]. Countries across the globe, and especially autocentric countries, can reduce their risk of missing GHG mitigation targets by reducing their dependence on private vehicles for road transportation alongside EV adoption.

Compact urban form (CUF) results in reduced road transportation, often measured as vehicle kilometers traveled (VKT), relative to sprawling urban form [12–19]. Additionally, CUF uses infrastructure such as roads, wastewater, drinking water, and electrical distribution more efficiently, reducing the resources and cost required per person to provide urban services [20,21]. CUF has been described both theoretically as walking and transit urban fabrics [22] and quantitatively through the optimization of the 'D variables'—density, diversity, design, destination accessibility, and distance to transit [14]. CUF reduces VKT by reducing trip length [13,14] and inducing modal shift from private transportation to transit and/or active transportation [23,24]. These trends in modal shift can become self-reinforcing; for example, safety perceptions can improve if more people use active transportation modes [25], while additional pedestrian footfall can attract local businesses, thereby increasing destination accessibility [26]. Additionally, transit operators may experience reduced operating deficits from higher patronage [27], enabling provision of higher frequency off-peak services.

CUF is more than population density; however, the population density of a whole settlement can be considered a coarse proxy for CUF [18,28]. Recently developed databases contain transportation and population density data on almost every settlement across the globe [29–32]. This near-census sample provides an opportunity to consider the universality of the relationship between population density and VKT observed in prior studies where sample size was constrained by laborious data collection, e.g., [18].

This paper will use these global databases [29–32] to develop the Compact Urban Form Estimation Tool (CUFET) for calculating the reduction in VKT and road transportation GHGs from shifting toward CUF. The predictive accuracy of the CUFET is assessed across economic output groups and over time, and the recommended use of the CUFET is constrained to where it makes the best predictions.

2. Materials and Methods

2.1. Model Data

Modeled road transportation VKT and GHGs from Climate TRACE [29,30] and settlement areas from the Global Human Settlement Layer (GHSL) [31,32] were used to develop the CUFET. The first year of road VKT and emissions factors in the Climate TRACE dataset was 2021 and the most recent year of settlement boundary delineation at the time of modeling was 2020 (GHSL updates boundaries every 5 years). Thus, 2021 VKT and 2021 emissions factors were used with 2020 settlement populations and boundaries. Population was obtained from the UN World Population Prospects 2022 estimates [31]. The modeled Climate TRACE VKT data for road segments was obtained from Google BigQuery and aggregated by GHSL delineated boundaries.

Detailed methods for describing how VKT and GHG emissions factors were calculated in the Climate TRACE dataset are described elsewhere [29] and will be briefly summarized here. The annual VKT of a settlement was calculated in the Climate TRACE dataset by summing the modeled VKT of all road segments by segment type (i.e., highway arterial, local) in a settlement area. The VKT of road segments was calculated as the capacity (total length of road network) multiplied by a capacity factor (average daily traffic for the year) which estimated the use of the road segment. The data for calculating road capacity and capacity factor came from open databases (e.g., OpenStreetMap), and data gaps were filled using satellite imagery (Sentinel-2) and machine learning algorithms.

GHG emissions were calculated by multiplying the VKT by a settlement-specific emissions factor that considered the combined 100-year GHG warming potential of CO_2, N_2O, and CH_4. Emissions factors were based on vehicle fleet mix, fuel type, and nominal fuel efficiency (adjusted by monthly temperatures). Emissions factors were calculated for each type of road segment for each municipality and multiplied by the activity of each road segment type. Calculated road transportation emissions were compared to the Electronic Data Gathering, Analysis, and Retrieval (EDGAR) v2025 international emissions inventory. Climate TRACE data compared well to the EDGAR values at mid- and high-emission settlements but contained a high amount of variability and predicted higher than actual emissions for low-emission settlements (<103 t CO_2eq from road transportation annually) [29].

Climate TRACE accounted for the % EV in their 2021 emissions factors [29]. However, as of 2021 EVs as a percent of global vehicle fleets was still only 1.4%, rising to 4.5% in 2024 [33]. Thus, the contribution of EVs is considered negligible for the 2021 emissions factors described here but need to be factored into future predictions of GHG emissions from ICE vehicles.

The GHSL settlement boundaries were determined algorithmically using population census and satellite imagery of the built environment. Settlement areas, termed Urban Centers by GHSL, were defined as "the spatially generalized high-density clusters of contiguous grid cells of 1 km^2 with a density of at least 1500 inhabitants per km^2 of land surface or at least 50% built-up surface share per km^2 of land surface, and a minimum population of 50,000" [31].

There were originally 11,260 settlement-level records in the 2021 Climate TRACE dataset. One settlement was removed because of satellite imagery errors (Roquetas de Mar in Spain, where satellite imagery mistakenly identified an extensive greenhouse operation as urban buildings). Four settlements were removed for insufficient temporal coverage (fewer than 12 months of data); 420 were removed for invalid VKT, population, or area values, resulting in 10,835 complete records across 188 countries. From this set, an additional 340 settlements from 89 small countries were removed because these countries had fewer than 10 settlements each, which would not allow for stable country fixed effects in the modeling framework. The final dataset used in the modeling analyses, therefore, consists of 10,495 settlements from 99 countries.

During the COVID-19 pandemic lockdowns in spring 2020, there were sharp decreases in road transport signaled by a drop in gasoline usage; however, gasoline consumption largely recovered by the fall of 2020 [34–36]. Personal correspondence with Climate TRACE engineers confirmed that impacts from COVID-19 disruptions to driving were largely resolved by 2021.

2.2. Modeling

Modeling and data diagnostic testing were conducted in Python. The VKT model was developed using settlement population, density, and a country fixed effect. Settlement population and area were highly correlated ($r_p = 0.857$) while population and density were poorly correlated ($r_p = 0.153$). Density was chosen over area to mitigate multicollinearity and improve the interpretability of model coefficients. VKT, population and density data were ln transformed to improve data normality, improve heteroskedasticity and allow for model coefficients to be interpreted as elasticities. A weighted least squares (WLS) model (1/population) was chosen as the Breusch–Pagan test indicated there was heteroskedasticity in the ordinary least squares (OLS) model with higher variability for lower population settlement areas. A comparison of the OLS and WLS models is shown in Table 1; however, heteroskedasticity persisted in the WLS model. Therefore MacKinnon–White (HC3) heteroskedasticity-robust standard errors are reported and used to calculate the statistical significance of model coefficients including country fixed effects. The country fixed effect is meant to capture all the variables that may influence road transportation VKT which were beyond the scope of the current modeling exercise. Population size banding at the 40th and 85th percentiles was incorporated into the model to improve the accuracy and capture the non-linear scaling effect of population and density on VKT. An 80/20 split of the data for training and testing was created using stratified random sampling by country and size bands to ensure proportional country and population size representation in both the training and testing subsets.

Table 1. Statistical diagnostic information of the OLS and CUFET VKT models.

Model	Set	Parameter	Value
OLS	Training	N	8396
		R^2	0.59
		F	6017 ($p < 0.001$)
		Durbin–Watson	2.040
		Variance Inflation Factor	1.0
		Breusch–Pagan (Lagrange Multiplier)	141.00 ($p < 0.001$)
		Cook's Distance ($D > 4\ n^{-1}$)	4.79%
		Cook's Distance ($D > 1$)	0%
CUFET (WLS + FE + Size Band)	Training	N	8396
		FE Countries	99
		R^2	0.68
		Adjusted R^2	0.67
		F	164.45 ($p < 0.001$)
		Durbin–Watson	2.035
		Variance Inflation Factor	1.0
		Breusch–Pagan (Lagrange Multiplier)	168.36 ($p < 0.001$)
		Cook's Distance ($D > 4\ n^{-1}$)	3.08%
		Cook's Distance ($D > 1$)	0%

Table 1. *Cont.*

Model	Set	Parameter	Value
CUFET (WLS + FE + Size Band)	Testing	N	2099
		R^2	0.78
		Aggregate Bias	+2.28%
		Root Mean Square Error (log-scale)	0.793

2.3. VKT Model, Economic Output Stratification and Backcasting

The predictive accuracy of the model was assessed using the percent difference in model-predicted VKT and measured VKT (calculated by Climate TRACE or Newman, Kenworthy, and Laube) for each settlement and year using the following equation:

$$Percent\ difference = \frac{2|p_i - m_i|}{p_i + m_i} \times 100\%$$

where:

p is the predicted value for record i;
m is the measured value for record i;
i is a record for settlement of a particular year (e.g., Toronto 2021).

The predictive accuracy of CUFET 2021 predictions were compared across countries grouped by deciles of 2021 GDP per capita (2021 USD) using World Bank data [37]. GDP data was unavailable for Cuba, North Korea, Palestine, South Sudan, Taiwan, and Yemen.

CUFET predictions were backcasted to compare the VKT data collected for settlements by Newman, Kenworthy, and Laube between 1960 and 2000 [18] and 1995/96 [38]. To make the temporal comparison consistent, all countries that were not present in the 1960 to 2000 dataset [18] were excluded from the 1995/96 dataset [38] from the backcasting exercise. The cities used in the backcasting exercise were from Australia, Belgium, Canada, Germany, Sweden, Switzerland and the United States. Differences in the modeled and measured VKT of the excluded cities will be briefly discussed in Section 4.2. The VKT of some cities from the Millenium Cities Database for Sustainable Transport were not modeled due to missing data (Buenos Aires, Brasilia, Salvador, Santiago, Turin, Casablanca, Lisbon, Istanbul) or because they had too few cities to produce a stable fixed effect for the CUFET model (Austria, Denmark, Finland, Greece, New Zealand, Norway, Singapore). The backcasting accuracy of the model for the remaining Millenium Cities Database for Sustainable Transport cities (n = 83) were compared to each country's GDP per capita. The average percent difference values were calculated for each year to see how the percent difference changed over time. A paired t-test was conducted for each year of the backcasting analyses.

2.4. GHG Calculation from VKT Model

Climate TRACE adjusted GHG emissions factors for each road segment type to account for the different driving patterns of different roads. The estimated GHG emissions from the simplified VKT prediction model used an emissions factor calculated for each settlement area. This settlement-specific emissions factor is applied to the predicted VKT value to estimate GHG emissions (Figure 1).

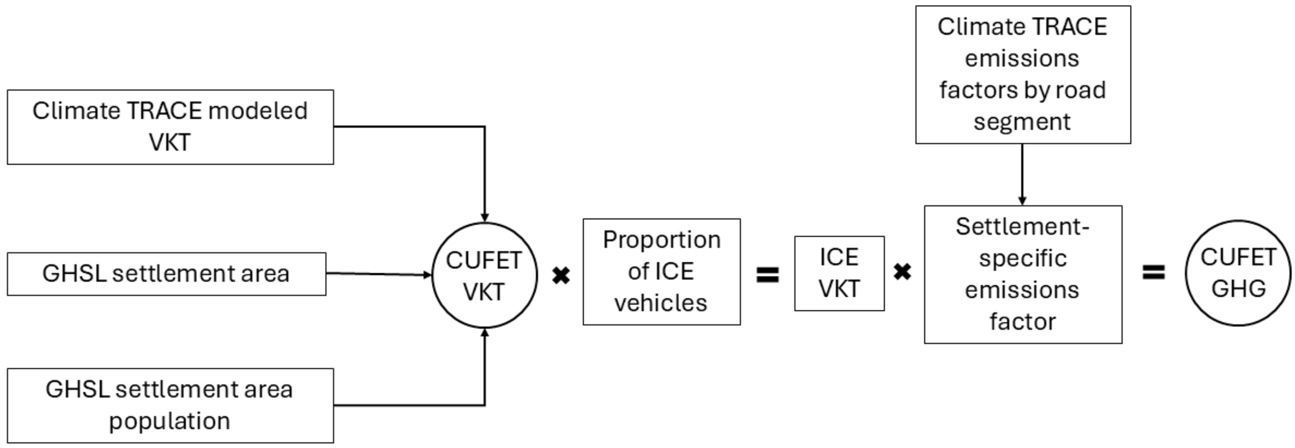

Figure 1. System diagram of CUFET model for settlement VKT and GHGs.

3. Results

3.1. VKT and GHG Emissions from Global Cities

Annual VKT and GHG emissions from the road transportation of global settlements relative to their population and density are shown in Figure 2. Settlements with densities that are relatively low for their populations have higher road transportation VKT and GHGs. The repeated linear breaks in points from lower population settlements are a relic of the 1km-by-1km grid used to define the GHSL boundaries.

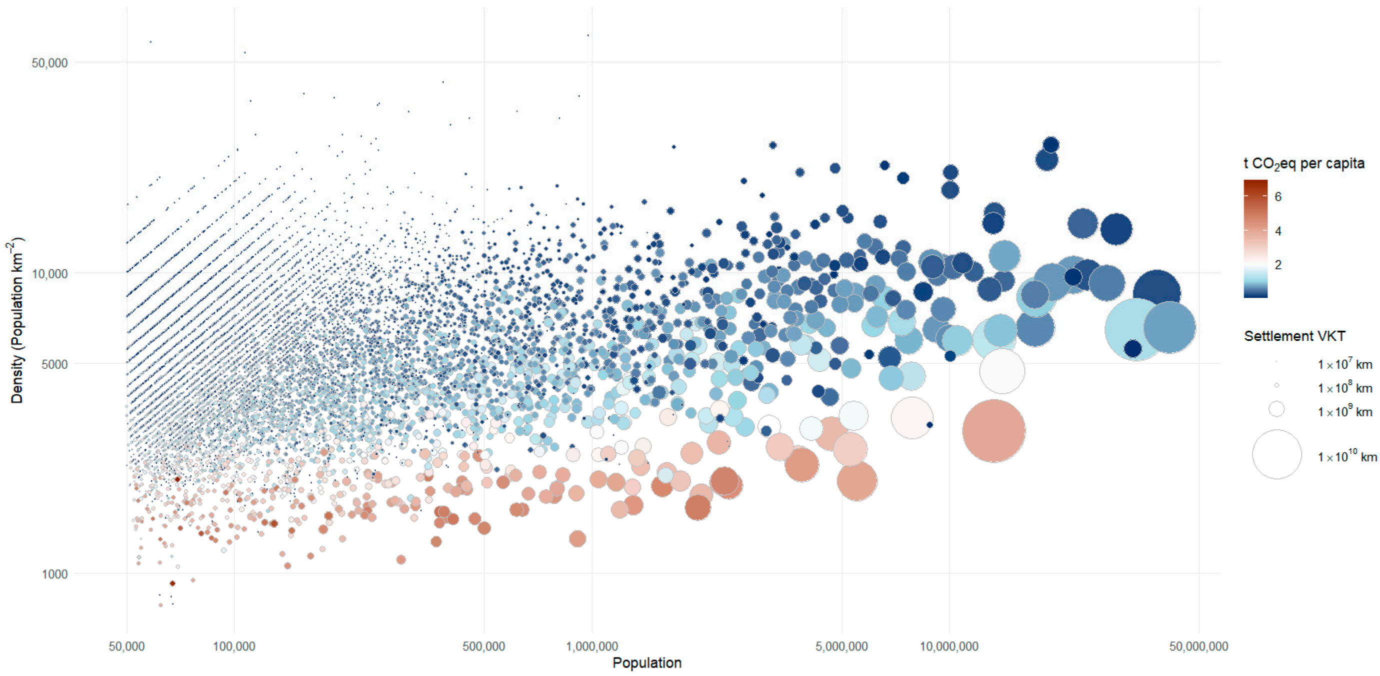

Figure 2. The vehicle kilometers traveled and GHG emitted from road transportation for 10,495 global settlements relative to their log density and log population. The size of the points is relative to the annual VKT of the entire settlement and the color of the points is related to the annual t CO_2eq per capita from road transportation. Emissions were calculated from the VKT using settlement-specific emissions factors. The points with 2 t CO_2eq annual emissions per capita are white; the points become increasingly more orange above 2 t CO_2eq and increasingly more blue below 2 t CO_2eq.

3.2. CUFET VKT Model

$$\ln(\mathrm{VKT_i}) = \beta_0 + \beta_1\cdot\ln(\mathrm{pop_i}) + \beta_2\cdot\ln(\mathrm{density_i}) + \gamma\cdot\mathrm{SizeBand} + \delta_k\cdot\mathrm{Country_k} + \beta_3\cdot\ln(\mathrm{pop}) \times \mathrm{SizeBand} + \beta_4\cdot\ln(\mathrm{density}) \times \mathrm{SizeBand} + \varepsilon_i$$

where:

β_0 is the y-intercept;
β_1 is the population elasticity;
β_2 is the density elasticity;
γ is the size band factor;
δ_k is the country fixed effect;
β_3 and β_4 are interaction terms that allow elasticities to vary by city size;
ε_i is an error term.

The predictive accuracy of the model improved from R^2 of 0.75 without size banding and to 0.78 (test data subset, n = 2099, aggregate bias = 2.3%, Root Mean Square Error = 0.793) with size banding, indicating that the associated impact of urban density and population growth on VKT was distinct at population thresholds. The OLS model using just population and density (with no fixed effect) had an R^2 of 0.59 for settlement VKT but with a much higher aggregate bias of 20%. The improvement in the explanatory power of the final CUFET model over the OLS model was largely due to the country fixed effect.

3.3. Model Elasticities and Country Fixed Effect

Population was a highly significant predictor of VKT for all size bands. The largest VKT elasticities with population growth were observed in small population settlements (1.58), followed by medium population settlements (1.28), with the smallest elasticities observed in large population settlements (1.04). Density was also a highly significant predictor of VKT for all size bands. The largest density elasticities were observed in small population settlements (−0.61), followed by medium (−0.51), with the smallest density elasticities in large population settlements (−0.21). This indicates that urban expansion (i.e., sprawl) has a proportionally larger effect on VKT in small and medium sized settlements (Table 2).

Table 2. Population and density elasticities, standard errors and significance for CUFET VKT model with small, medium and large population settlements.

Parameter	Size Band	Elasticity	SE (HC3)	*p*-Value
Population (β_1)	Small (50,000 to 88,335)	1.583	0.110	<0.001
	Medium (88,335 to 329,480)	1.283	0.058	<0.001
	Large (>329,480)	1.036	0.039	<0.001
Density (β_2)	Small (50,000 to 88,335)	−0.605	0.079	<0.001
	Medium (88,335 to 329,480)	−0.512	0.079	<0.001
	Large (>329,480)	−0.206	0.077	0.007

Country fixed effects were statistically significant ($p < 0.05$) in 86 of the 99 tested, confirming substantial country-level variation in road transportation VKT independent of density or population size. The countries' fixed effects factors had a range of 3.9, from the lowest value −1.18 with Bangladesh to the highest value of 2.72 with the United States. Afghanistan (first country in alphabetical order) was the arbitrarily chosen reference value at 0.

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3.4. VKT Model, Economic Output Stratification and Backcasting

The predictive performance as measured by the % difference in modeled VKT vs. VKT measured in the full Climate TRACE dataset was compared across national per capita economic output (Table 3). The CUFET model achieves an average of 23.9% and 19.0% difference in the ninth and tenth decile respectively but the predictive power of the model drops off with lower economic output countries. The tenth decile contains the countries (with average % differences) Australia (20.6%), Belgium (20.6%), Canada (16.2%), Germany (21.1%), Israel (30.7%), the Netherlands (16.1%), Sweden (20.7%), Switzerland (17.0%), United Kingdom (19.8%), and the United States (17.9%). The ninth decile contains the countries (with average % differences) Czechia (13.6%), France (19.3%), Hungary (15.0%), Italy (21.4%), Japan (20.2%), Saudi Arabia (31.9%), South Korea (29.5%), Spain (24.6%), and the United Arab Emirates (68.3%).

Table 3. Percent difference in modeled vs. measured settlement VKT stratified by 2021 country GDP per capita decile.

Country 2021 GDP per Capita Decile	Number of Settlements	Decile Range GDP per Capita (2021 USD)	Mean % Difference
Data Unavailable	218	NA	62.4
1	494	356–885	74.1
2	455	893–1243	50.4
3	534	1245–2061	49.0
4	2302	2138–2735	82.4
5	992	2787–4183	50.6
6	784	4287–6061	40.1
7	807	6223–9982	30.0
8	2645	9984–18,636	40.9
9	509	19,031–44,119	23.9
10	755	47,691–93,665	19.0

Data collected by Newman, Kenworthy, and Laube on the VKT of cities from 1960 to 2000 [18] and 1995/1996 [38] was used to (1) check if the model would give reasonable estimates using input data based on administrative boundaries and (2) use past backcasting performance to constrain the forecasting horizon (Figure 3, Table 4). The model had a larger mean % difference in 1960 (82%) than the unexplained error in the 2021 test sample (22%), but became increasingly more predictive each decade and leveled off around 20% after 1990. In 1990, 1995/96 and 2000 the model had a similar percent difference to the unexplained error in the 2021 test sample (Table 4) and the model predictions in 1990, 1995/96 and 2000 were not significantly different from the measured results. The poor performance of the model in 1960 is to be expected as the country fixed effect would be divorced from the 2021 context in which it was calibrated. The backcasting suggests that the fixed effect for the CUFET VKT may be useful for estimates within 30 years of the modeling date.

Table 4. A summary of the percent difference in settlement VKT measured by Newman, Kenworthy, and Laube [18,38] and settlement VKT predicted by the CUFET by year.

Year	Number of Settlements	Mean % Difference	Paired t	*p*-Value
1960	18	82.0	−2.37	0.030
1970	17	53.9	−2.31	0.034
1980	24	35.1	−2.39	0.025
1990	24	20.0	−1.66	0.110

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Table 4. *Cont.*

Year	Number of Settlements	Mean % Difference	Paired t	*p*-Value
1995/96	31	16.1	−0.66	0.512
2000	24	20.1	−1.02	0.320

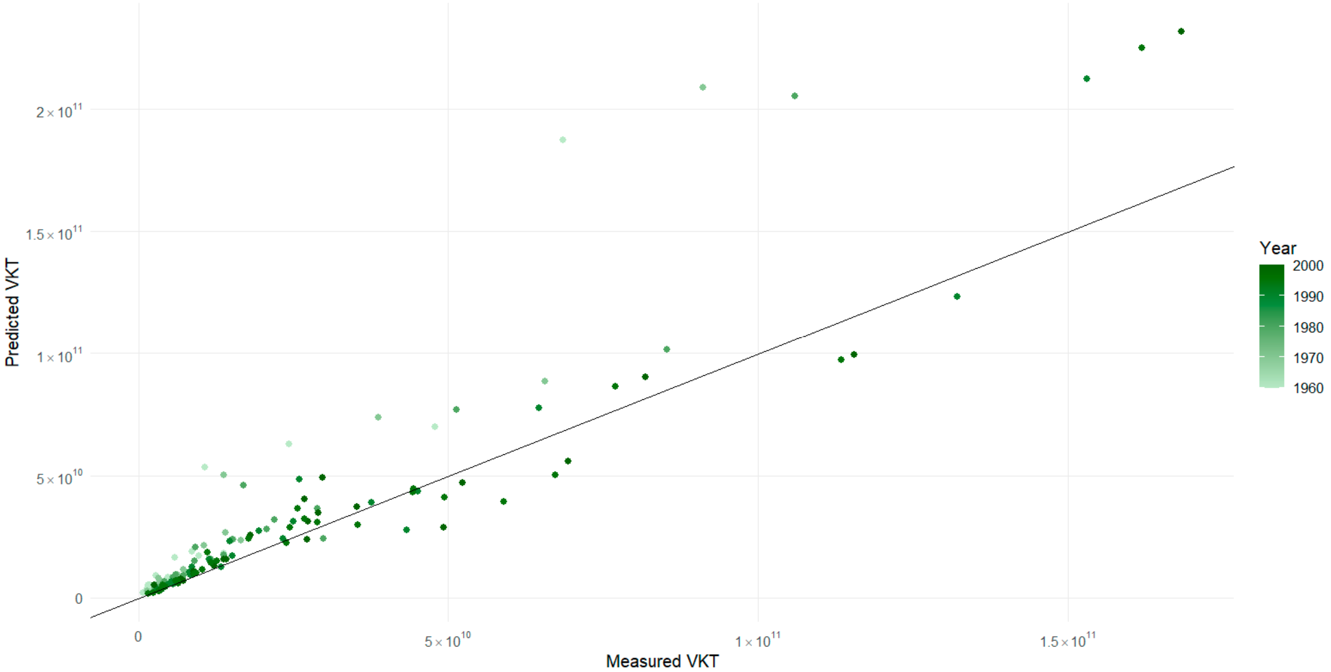

Figure 3. **The** VKT measured by Newman, Kenworthy, and Laube [18,38] vs. VKT predicted by the CUFET with a 1:1 line.

4. Discussion

4.1. CUFET Strengths and Limitations

This study affords a big picture quantitative approach for estimating VKT reductions associated with urban densification. The imperative to achieve substantive decarbonization highlights the importance of removing barriers to the global sustainability transition across key sectors such as transportation. This paper's contribution is to help empower decision-makers seeking to accelerate a transition to sustainable transportation in high economic output nations, where it is most needed, by providing a cost-efficient model to estimate transportation GHG emission reductions from CUF.

The CUFET model captures a near census of global settlements and removes selection bias errors, and by using the GHSL instead of administrative boundaries, the CUFET avoids the impacts of arbitrary scale [39,40], solving the urban boundary problem with a consistent definition for all settlements across the globe [41]. The CUFET model captures VKT and GHG emissions from exurban and suburban transportation in sprawling urban metropolises that may be missed by administrative boundaries. However, backcasting performance using administrative boundaries suggests the model is robust to the urban boundary problem.

The limitations of the CUFET model are that it uses compound variables for both country fixed effects in the VKT model and settlement-specific emission factors when converting VKT to GHG emissions. These compound variables are aggregates of several distinct variables. For example, the settlement-specific emission factor captures the variation in vehicle fleet mix, fuel type, and nominal fuel efficiencies [29]. This estimation model is best interpreted as determining what difference in VKT and road transportation GHG emissions will likely be observed if density and population changes, with 'all else being equal'. There is, of course, potential useful GHG mitigation that could occur from a shift to more fuel-efficient smaller vehicles, for example; however, those considerations are outside of the scope of the current modeling exercise.

The current iteration of CUFET is less accurate for smaller population cities (Table 2) and limited to making reasonable predictions for higher economic output countries (Table 3). As Australia, Canada, the United States and Japan have the greatest need to shift their autocentric settlements toward compact urban from, the CUFET can be a useful tool in these countries. Future iterations of the CUFET should expand the effective scope of model predictions to be useful in assisting lower economic output countries assess the benefits of CUF.

Additionally, it is acknowledged that the presented methodology does not account for the rich variety of context-specific mechanisms of real-world urban densification. Hence, while this model can readily generate plausible and transparent future VKT estimates associated with CUF, it is not the authors' intent to discount the importance of place-based factors that may alternatively increase or decrease the influence of urban densification as a key independent variable.

4.2. VKT Model, Economic Output Stratification and Backcasting

CUFET can provide reasonable estimates of VKT with changes in population and density for higher economic output countries (>$20,000 per capita in 2021 USD). The high economic output countries with the largest % difference in model performance were all outside of North America and Europe (Israel, Saudi Arabia, South Korea and the United Arab Emirates). The stratification in predictive performance related to economic output may be related to a threshold below which many people who want cars are not able to afford them.

Backcasting used administrative boundaries [18,38] to calculate density, not the systemically derived settlement areas from the GHSL [31], which introduces unaccounted-for error in model estimates. Despite this the model backcasting performance from 1990 onwards was as accurate for high economic output countries as using the 2021 Climate TRACE dataset the model was trained and tested on. Past backcasting performance indicates that the model makes reasonable estimates 30 years prior to model calibration data from 2021. However, the forecasting accuracy of the CUFET remains uncertain.

The large deviations of the six points from the 1:1 line near the top of Figure 3 are all New York for each year data was available from 1960 to 2000. The fixed effect for the United States does a poor job predicting VKT in New York as New York has distinct transportation patterns compared to other United States cities [42]. If the fixed effect for New York is adjusted from the US fixed effect of 2.71 to the UK fixed effect of 2.39, it reduces the % difference in all the points by 30% and brings the 1990 and 2000 VKT model predictions within 1% of the measured VKT. The example of New York highlights the sensitivity of the model's predictions on the country fixed effect. Practitioners of the model may wish to use a backcasting exercise to calibrate a more accurate fixed effect for their city.

A total of 52 cities were excluded from the backcasting exercise as they only had data for a single year (1995/96). Many of these cities had a % difference in the modeled VKT that was higher than the countries included in the backcasting analyses with a mean difference of 53.8%. The largest differences were found in Dakar, Senegal (170.1%); Cairo, Egypt (133.3%); and Bogota, Colombia (126.6%). If we take the full modeled set 1995/96 Millenium Cities Database for Sustainable Transport (n = 83) and stratify it by 2021 GPD per capita [37] there is a similar divergence in backcasting performance as with the 2021 data at \$20,000 per capita USD. For countries above \$20,000 per capita (n = 53) the mean % difference in model backcasting was 25.2%; for countries below \$20,000 per capita (n = 30) the mean % difference in model backcasting was 67.1%. This further supports that the CUFET model is more effective at modeling VKT in higher economic output countries.

4.3. Transportation GHG Emissions and Density

Several institutions have highlighted 2–2.5 t CO_2eq per capita per year as a near-term (e.g., by 2030) target carbon budget for citizens of wealthy countries to avoid the worst impact of Climate Change [43–45]. Figure 2 only shows modeled transportation GHG emissions, not total GHG emissions. Road transportation emissions above 2 t CO_2eq per capita are incompatible with the goal of avoiding the worst impacts of Climate Change and why a color break of 2 t CO_2eq per capita was used in Figure 2.

The data in Figure 2 shows a pattern consistent with the literature: settlements with lower density at a given population tend to have higher annual road transportation GHG emissions relative to more compact settlements [12–19]. Of the 437 settlements that had road transportation GHG emissions per capita above 2 tonnes annually, 312 of them were in the United States, 68 were in Japan, 24 were in Canada and 21 were in Australia. Settlements in these countries have the largest opportunity to reduce VKT and transportation emissions from transitioning towards compact urban form.

Japan has a reputation for efficient transportation and is reported to have lower transportation emissions than OECD peer countries [46]. Japan's most populus cities, Tokyo and Osaka, had transportation emissions of 1.15 and 1.35 t CO_2eq per capita respectively in 2021. However, many smaller Japanese cities, such as Hamamatsu and Tosu, are more autocentric and had transportation emissions of 4.51 and 3.00 t CO_2eq per capita respectively in 2021.

4.4. Practical Use of CUFET

The CUFET is a simple VKT prediction model and can be used to estimate the changes in VKT, in cites of high economic output countries (>\$20,000 per capita in 2021 USD) for different growth scenarios by using population change projections and comparing different density scenarios. Additionally, the percent of VKT from electric vehicles can easily be adjusted to allow for estimates that consider a variety of EV uptake scenarios.

Low settlement density is not a direct cause of road transportation emissions, but instead a proxy for causal factors (e.g., automotive urban fabric or poor D-variable values) that result in a higher proportion of private vehicle transportation, less transit, and less active transport. Users of the model should avoid the naive interpretation that adding settlement-wide density with no other consideration of transit and active transportation will be sufficient to effectively reduce road transportation emissions. Practical guides such as LEED v4.1 Cities and Communities [47] should be used when implementing compact urban form.

The CUFET model allows estimates of GHG reduction from CUF that can be computed with minimal cost and effort. The CUFET is not intended to replace more complex and accurate transportation models but rather to be used to help prioritize municipal GHG-mitigating initiatives such as more sophisticated transportation modeling that would capture details such as transit service, bike lanes, land-use mix and vehicle fleet composition. Additionally, the simplicity of the CUFET will allow lay users and students to interact with the CUFET model directly, allowing the model to be used as a communication and teaching tool. A spread sheet populated with CUFET model parameters (S1) can be found in the Supplementary Materials of this publication. The model allows you to enter the percent of EVs such that the transportation GHG reduction from CUF can be estimated independently from EVs.

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4.5. Considerations for Further Research

Although the model can generate estimates of future VKT and transportation GHG emissions across different future urban density scenarios, it cannot substitute the need for further research and consultation by decision-makers such as local municipalities or regional planning bodies. The model is not intended to provide definitive answers relating to sustainable 'end states' in terms of the necessary thresholds or distributions of density and cannot in isolation from other approaches clarify the many policy trade-offs and unintended consequences inherent to the nature of planning.

Beyond the design of the model, substantive questions remain about the likely velocity of change and the ability of spatial planning to contribute to timely decarbonization [48]. Large swathes of peri-urban land have already developed which are arguably resistant to substantive change due to their location, structure and institutional path dependency [22]. We do not know how much VKT reduction is possible through CUF policy reform as there are uncertainties about the plausible rate of urban change.

Further research could explore disaggregating the country fixed effect to improve the robustness of the CUFET and expand its applicability beyond countries with GDP per capita greater than $20,000 (2021 UDS). Improved predictive power could be achieved by including economic information as a model input variable. Further disaggregating the country fixed effect could be investigated by looking for relationships of fixed effect coefficients with more detailed information within the GHSL datasets and other global datasets. There may also be opportunities to develop finer scale fixed affects for larger countries such as China, India and the United States. Any additional inclusion of a predictor variable would need to be weighed against the cost of inclusion to the end user as the primary use case for the CUFET is to provide estimates with minimal effort.

Supplementary Materials: The following supporting information can be downloaded at https://www.mdpi.com/article/10.3390/app16125828/s1, Spreadsheet S1: CUFET.xlsx.

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Abbreviations

The following abbreviations are used in this manuscript:

CUF	Compact urban form
CUFET	Compact Urban Form Estimation Tool
EDGAR	Electronic Data Gathering, Analysis, and Retrieval
EV	Electric vehicle
GHG	Greenhouse gas
GHSL	Global Human Settlement Layer
GDP	Gross Domestic Product
ICE	Internal combustion engine
LEED	Leadership in Energy and Environmental Design
OLS	ordinary least squares
UN	United Nations
VKT	Vehicle kilometers traveled
WLS	weighted least squares

References

1. Olhoff, A.; Lamb, W.; Kuramochi, T.; Rogelj, J.; den Elzen, M.; Christensen, J.; Fransen, T.; Pathak, M.; Tong, D. (Eds.) *Emissions Gap Report 2025: Off Target—Continued Collective Inaction Puts Global Temperature Goal at Risk*; United Nations Environment Programme: Nairobi, Kenya, 2025. [CrossRef]
2. Crippa, M.; Guizzardi, D.; Pagani, F.; Banja, M.; Muntean, M.; Schaaf, E.; Monforti-Ferrario, F.; Becker, W.; Quadrelli, R.; Risquez Martin, A.; et al. *GHG Emissions of All World Countries*; Publications Office of the European Union: Luxembourg, 2024. [CrossRef]
3. Kenworthy, J.R.; Laube, F.B. Patterns of Automobile Dependence in Cities: An International Overview of Key Physical and Economic Dimensions with Some Implications for Urban Policy. *Transp. Res. Part A Policy Pract.* **1999**, *33*, 691–723. [CrossRef]
4. Milovanoff, A.; Posen, I.D.; MacLean, H.L. Electrification of Light-Duty Vehicle Fleet Alone Will Not Meet Mitigation Targets. *Nat. Clim. Change* **2020**, *10*, 1102–1107. [CrossRef]
5. Li, Y.; Jenn, A. Impact of Electric Vehicle Charging Demand on Power Distribution Grid Congestion. *Proc. Natl. Acad. Sci. USA* **2024**, *121*, e2317599121. [CrossRef] [PubMed]
6. Labriet, M.; Joshi, S.R.; Vielle, M.; Holden, P.B.; Edwards, N.R.; Kanudia, A.; Loulou, R.; Babonneau, F. Worldwide Impacts of Climate Change on Energy for Heating and Cooling. *Mitig. Adapt. Strateg. Glob. Change* **2015**, *20*, 1111–1136. [CrossRef]
7. Nøland, K.J.; Hjelmeland, M.; Korpås, M. Will Energy-Hungry AI Create a Baseload Power Demand Boom? *IEEE Access* **2024**, *12*, 110353–110360. [CrossRef]
8. Von Hippel, T. Thermal Removal of Carbon Dioxide from the Atmosphere: Energy Requirements and Scaling Issues. *Clim. Change* **2018**, *148*, 491–501. [CrossRef]
9. Jones, B.; Elliott, R.J.R.; Nguyen-Tien, V. The EV Revolution: The Road Ahead for Critical Raw Materials Demand. *Appl. Energy* **2020**, *280*, 115072. [CrossRef]
10. Söltzer, L.; Wortmann, B.; Stolten, D.; Linßen, J.; Heinrichs, H. Material Bottlenecks of Batteries within the Energy Transition. *Resour. Conserv. Recycl.* **2026**, *227*, 108745. [CrossRef]
11. Woodley, L.; See, C.Y.; Cook, P.; Yeo, M.; Palmer, D.S.; Huh, L.; Wang, S.; Nunes, A. Climate Impacts of Critical Mineral Supply Chain Bottlenecks for Electric Vehicle Deployment. *Nat. Commun.* **2024**, *15*, 6813. [CrossRef]
12. Bart, I.L. Urban Sprawl and Climate Change: A Statistical Exploration of Cause and Effect, with Policy Options for the EU. *Land Use Policy* **2010**, *27*, 283–292. [CrossRef]
13. Brownstone, D.; Golob, T.F. The Impact of Residential Density on Vehicle Usage and Energy Consumption. *J. Urban Econ.* **2009**, *65*, 91–98. [CrossRef]
14. Ewing, R.; Cervero, R. Travel and the Built Environment: A Meta-Analysis. *J. Am. Plan. Assoc.* **2010**, *76*, 265–294. [CrossRef]
15. Jones, C.; Kammen, D.M. Spatial Distribution of U.S. Household Carbon Footprints Reveals Suburbanization Undermines Greenhouse Gas Benefits of Urban Population Density. *Environ. Sci. Technol.* **2014**, *48*, 895–902. [CrossRef] [PubMed]
16. Larson, W.; Liu, F.; Yezer, A. Energy Footprint of the City: Effects of Urban Land Use and Transportation Policies. *J. Urban Econ.* **2012**, *72*, 147–159. [CrossRef]
17. Newman, P.W.G.; Kenworthy, J.R. Transport and Urban Form in Thirty-two of the World's Principal Cities. *Transp. Rev.* **1991**, *11*, 249–272. [CrossRef]
18. Newman, P.; Kenworthy, J. *The End of Automobile Dependence: How Cities Are Moving Beyond Car-Based Planning*; Island Press/Center for Resource Economics: Washington, DC, USA, 2015. [CrossRef]

19. Norman, J.; MacLean, H.L.; Kennedy, C.A. Comparing High and Low Residential Density: Life-Cycle Analysis of Energy Use and Greenhouse Gas Emissions. *J. Urban Plan. Dev.* **2006**, *132*, 10–21. [CrossRef]
20. Carruthers, J.I.; Ulfarsson, G.F. Urban Sprawl and the Cost of Public Services. *Environ. Plan. B Plan. Des.* **2003**, *30*, 503–522. [CrossRef]
21. Müller, D.B.; Liu, G.; Løvik, A.N.; Modaresi, R.; Pauliuk, S.; Steinhoff, F.S.; Brattebø, H. Carbon Emissions of Infrastructure Development. *Environ. Sci. Technol.* **2013**, *47*, 11739–11746. [CrossRef] [PubMed]
22. Newman, P.; Kosonen, L.; Kenworthy, J. Theory of Urban Fabrics: Planning the Walking, Transit/Public Transport and Automobile/Motor Car Cities for Reduced Car Dependency. *Town Plan. Rev.* **2016**, *87*, 429–458. [CrossRef]
23. Cervero, R. Mixed Land-Uses and Commuting: Evidence from the American Housing Survey. *Transp. Res. Part A Policy Pract.* **1996**, *30*, 361–377. [CrossRef]
24. Gössling, S.; Schröder, M.; Späth, P.; Freytag, T. Urban Space Distribution and Sustainable Transport. *Transp. Rev.* **2016**, *36*, 659–679. [CrossRef]
25. Hwang, U.; Guhathakurta, S. Exploring the Impact of Bike Lanes on Transportation Mode Choice: A Simulation-Based, Route-Level Impact Analysis. *Sustain. Cities Soc.* **2023**, *89*, 104318. [CrossRef]
26. Volker, J.M.B.; Handy, S. Economic Impacts on Local Businesses of Investments in Bicycle and Pedestrian Infrastructure: A Review of the Evidence. *Transp. Rev.* **2021**, *41*, 401–431. [CrossRef]
27. Moore, G. Transit Ridership Efficiency as a Function of Fares. *J. Public Transp.* **2002**, *5*, 105–131. [CrossRef]
28. Ewing, R.; Cervero, R. "Does Compact Development Make People Drive Less?" The Answer Is Yes. *J. Am. Plan. Assoc.* **2017**, *83*, 19–25. [CrossRef]
29. Foster, K.; Sicurello, P.; Piatko, C.; Reilly, E. *Transportation Sector: Global Road Emissions*; The Johns Hopkins University Applied Physics Laboratory (JHU/APL): Laurel, MD, USA, 2025; Climate TRACE Emissions Inventory; Version Fall 2025; Climate TRACE.
30. Kott, T.; Foster, K.; Villafane-Delgado, M.; Loschen, W.; Sicurello, P.; Ghebreselassie, M.; Reilly, E.; Hughes, M. Transportation Sector—Global Road Emissions; The Johns Hopkins University Applied Physics Laboratory (JHU/APL), USA, Climate TRACE Emissions Inventory. 2024. Available online: https://climatetrace.org (accessed on 28 November 2025).
31. Pesaresi, M.; Schiavina, M.; Politis, P.; Freire, S.; Krasnodębska, K.; Uhl, J.H.; Carioli, A.; Corbane, C.; Dijkstra, L.; Florio, P.; et al. Advances on the Global Human Settlement Layer by Joint Assessment of Earth Observation and Population Survey Data. *Int. J. Digit. Earth* **2024**, *17*, 2390454. [CrossRef]
32. Freire, S.; Schiavina, M.; Corbane, C.; Pesaresi, M.; Sabo, F.; Tommasi, P.; Airaghi, D.; Ehrlich, D.; Melchiorri, M.; Politis, P.; et al. *GHS-UCDB R2019A—GHS Urban Centre Database 2015, Multitemporal and Multidimensional Attributes*; European Commission, Joint Research Centre: Brussels, Belgium, 2019. [CrossRef]
33. International Energy Agency. *Global EV Data; Global EV Outlook 2025*; Global EV Data Dataset; International Energy Agency: Paris, France, 2025. Available online: https://www.iea.org/data-and-statistics/data-product/global-ev-outlook-2025#global-ev-data (accessed on 5 March 2026).
34. Naderi, S.; Tian, X.; An, C. Assessment of Mobility Trends and Transportation-Related Emissions in Canadian Cities during the Post-COVID-19 Pandemic Period. *Environ. Syst. Res.* **2025**, *14*, 3. [CrossRef]
35. Ou, S.; He, X.; Ji, W.; Chen, W.; Sui, L.; Gan, Y.; Lu, Z.; Lin, Z.; Deng, S.; Przesmitzki, S.; et al. Machine Learning Model to Project the Impact of COVID-19 on US Motor Gasoline Demand. *Nat. Energy* **2020**, *5*, 666–673. [CrossRef]
36. Patino-Artaza, H.; King, L.C.; Savin, I. Did COVID-19 Really Change Our Lifestyles? Evidence from Transport Energy Consumption in Europe. *Energy Policy* **2024**, *191*, 114204. [CrossRef]
37. World Bank Group. World Bank Open Data, GDP per Capita (Current US$). Available online: https://data.worldbank.org/indicator/NY.GDP.PCAP.CD (accessed on 16 February 2026).
38. Kenworthy, J.; Laube, F. Millenium Cities Database for Sustainable Mobility. International Union of Public Transport. Available online: https://trid.trb.org/View/708144 (accessed on 27 March 2026).
39. Arcaute, E.; Hatna, E.; Ferguson, P.; Youn, H.; Johansson, A.; Batty, M. Constructing Cities, Deconstructing Scaling Laws. *J. R. Soc. Interface* **2015**, *12*, 20140745. [CrossRef]
40. Cottineau, C.; Hatna, E.; Arcaute, E.; Batty, M. Diverse Cities or the Systematic Paradox of Urban Scaling Laws. *Comput. Environ. Urban Syst.* **2017**, *63*, 80–94. [CrossRef]
41. Oliveira, E.A.; Furtado, V.; Andrade, J.S.; Makse, H.A. A Worldwide Model for Boundaries of Urban Settlements. *R. Soc. Open Sci.* **2018**, *5*, 180468. [CrossRef] [PubMed]
42. Pan, M.; Patwary, L.; Reuscher, T.; Uddin, M. *New York State Household Travel Patterns: A Comprehensive Comparison Analysis*; ORNL-SPR–2025-3769, 3002637; Oak Ridge National Laboratory (ORNL): Oak Ridge, TN, USA, 2025. [CrossRef]
43. Akenji, L.; Bengtsson, M.; Toivio, V.; Lettenmeier, M.; Fawcett, T.; Parag, T.; Saheb, Y.; Coote, A.; Spangenberg, J.H.; Capstick, S.; et al. *1.5-Degree Lifestyles: Towards a Fair Consumption Space for All*; Hot or Cool Institute: Berlin, Germany, 2021.
44. Gore, T. *Carbon Inequality in 2030: Per Capita Consumption Emissions and the 1.5 °C Goal*; Institute for European Environmental Policy: London, UK; Oxfam: Oxford, UK, 2021. [CrossRef]

45. TUDelft. *2 Tons 2 Thrive*; Delft University of Technology: Delft, The Netherlands, 2025. Available online: https://www.tudelft.nl/sustainability/environmental-literacy/2t2t (accessed on 25 November 2025).
46. OECD. *OECD Environmental Performance Reviews: Japan 2025*; OECD Environmental Performance Reviews; OECD Publishing: Paris, France, 2025. [CrossRef]
47. U.S. Green Building Council. *LEED v4.1*; U.S. Green Building Council: Washington, DC, USA, 2019. Available online: https://www.usgbc.org/leed/v41 (accessed on 19 February 2026).
48. Moos, M. The Velocity of Density: Can We Build More Sustainable Cities Fast Enough? *Sustainability* **2017**, *9*, 2326. [CrossRef]